

## Matched successful goals: paired results

# Public xArm matched-goal diagnostic

Six successful episodes, 30 paired timestamps. Each episode now uses its own final successful frame as its visual goal. This is a hindsight diagnostic, not a deployable reference-selection rule.

Primary analysis excludes the exact terminal goal image in BOTH conditions: indices 4, 15, 31, 47. Native score definitions and every observed frame/history are unchanged. Higher VOC means a stronger linear association with time along these successful trajectories.

| Method                         | Independent VOC | Matched VOC | Paired change | 95% paired task-bootstrap interval for change |
|--------------------------------|-----------------|-------------|---------------|-----------------------------------------------|
| PSI DT                         | 0.061           | 0.972       | +0.911        | [+0.365, +1.443]                              |
| PSI S1                         | 0.149           | 0.857       | +0.709        | [+0.209, +1.311]                              |
| PSI S3                         | 0.276           | 0.860       | +0.584        | [+0.118, +1.183]                              |
| PSI S5                         | 0.286           | 0.864       | +0.578        | [+0.125, +1.167]                              |
| DINOv2                         | 0.544           | 0.501       | -0.044        | [-0.915, +0.838]                              |
| Robometer progress (unchanged) | 0.933           | 0.933       | +0.000        | [+0.000, +0.000]                              |
| Encoded time                   | 1.000           | 1.000       | +0.000        | [+0.000, +0.000]                              |

Secondary analysis includes terminal frame 63 and is expected to benefit from endpoint identity. All five-prefix metrics and individual tasks are in the downloadable CSV.

Only six independent task/episode units; bootstrap intervals are exploratory. Four samples per episode make correlations unstable. The goal contains future scene and arm-pose information; matching can improve appearance similarity without establishing semantic progress recognition.

The 24 failed/suboptimal attempts have no verified matched successful targets in this release. Their endpoints are NOT used as successful goals. Outcome AUROC and terminal ranking are therefore not recomputed. Existing independent-goal outcome results remain unchanged.

Robometer is unchanged because it consumes language rather than the goal image. Encoded time has VOC 1 by definition; successful-only VOC cannot test failure recognition. Public videos are resampled to 64 frames and do not retain original time, so DT is not interpreted as seconds.

Reference audit: ../MATCHED\_REFERENCE\_AUDIT.md. Source pins, checkpoint and tokenization are inherited from ../xarm\_v1/run\_receipt.json.

# Public reference availability audit

## # Public matched-success reference audit

Inspected pinned USC xArm raw release, all 36 annotated videos/first frames, 38 total MP4 files, README, parquet, Arrow, state/dataset-info metadata, and official upload loader.

No scene IDs, reset IDs, per-attempt paired-success IDs, independent goal images, original source-folder IDs, or simulator states are exported. The loader parses attempt IDs but discards them, generating random UUIDs. All partial-success fields are null. Videos are real robot recordings; ABC simulator assets cannot reconstruct their physical resets.

There are two successful demonstrations per task, but all 24 failed/suboptimal attempts lack a verified same-reset success pairing. First-frame comparisons give 0/48 exact pixel matches; this is only a duplicate diagnostic, not a physical-scene proof. Visual inspection independently shows significant layout variation for cup/banana/eggplant tasks. Drawer/stacking fixtures look similar, but exact pairing remains unverified.

Two unannotated MP4s are exact duplicates of existing successful clips. A 38-row Arrow export has different UUIDs and shifted quality labels and supplies no pairing fields. The repository README selects the 36-row train-\*.parquet used in the pilot; the Arrow export must not be merged as additional reference data.

Available next step: the six evaluated successful episodes can each use their own final image in a clearly labeled matched hindsight diagnostic, preferably excluding exact terminal targets from primary analysis. A fully matched success-versus-incomplete benchmark needs new paired physical recordings, original reset/pair metadata from the collectors, or a separately labeled constructed-goal experiment. No failure endpoint should be treated as its successful goal.

Audit receipt: `matched_reference_audit.json`; visual evidence: `initial_scene_reference_audit.jpg`. No model scores or episode labels were changed.